

Sign Language Video-to-Text Translation with Reinforcement Learning: Spatio-Temporal Visual Encoders, Decoder Fine-Tuning Algorithms and Reward Design on PHOENIX-2014T and How2Sign

Van Linh Nguyen¹

Abstract

Sign Language Translation (SLT) maps a continuous video of a signer to a natural-language sentence, and is a video-understanding problem before it is a language problem: the input is a spatio-temporal visual signal that must be detected, tracked, temporally resampled and encoded before a single token can be emitted. This work therefore sits squarely in image and video processing, and treats the visual front-end as a first-class experimental axis rather than as fixed preprocessing. We present a gloss-free SLT pipeline whose visual stage extracts a 183-d per-frame skeletal representation with MediaPipe Holistic — per-frame landmark regression, per-segment interpolation of missed detections, and uniform temporal resampling to at most 300 frames — and whose translation stage is warm-started with cross-entropy and then fine-tuned by Reinforcement Learning (RL) directly toward the evaluation metric, addressing the exposure bias and loss-metric mismatch of teacher forcing. On the vision axis we compare six interchangeable spatio-temporal encoders behind a single interface (Transformer, GCN, ST-GCN, Graph Transformer, dilated TCN, Perceiver IO), spanning global attention, convolutional temporal locality and anatomical graph priors, and we report quality against both parameter count and inference latency. On the RL axis we compare eight decoder fine-tuning algorithms and variants (SCST, REINFORCE, PPO, A2C, MRT, RAML, DPO, Curriculum RL) under a single multi-objective reward, plus a reward ablation separating BLEU gains from genuine quality (repetition and length). Every experiment is reported relative to mandatory model-free floor baselines. To fit a free-hardware budget (Kaggle T4 x2), the study is organised around three training-subset sizes (5, 10, and 25% of the PHOENIX-2014T training split, with development and test sets always evaluated in full), forming a data-size ablation; a How2Sign trial then varies both the sign language and the visual modality, contrasting RGB-derived I3D video features against skeletal pose features at 10, 25 and 50%. Three findings stand out. First, the visual representation dominates: RGB-derived I3D features beat

body pose at every How2Sign subset (**4.25** against **1.62** BLEU-4 at 50%) and only I3D scales with data, making the pose-only constraint a measurable ceiling rather than a free design choice. Second, the encoder ranking inverts with scale, with graph encoders overtaking the plain Transformer at 25% and the lightest ST-GCN reaching higher quality with 23% fewer parameters and a quarter of the latency. Third, RL decoder fine-tuning gives no reliable gain at the smallest 5% subset (the SCST advantage is smaller than run-to-run variance) but a genuine, growing gain at 10 and 25%, led by the multi-sample estimator MRT (up to **+0.76** BLEU-4); on How2Sign, whose target sentences are markedly longer, the sequence-level objective is what keeps improving after the token-level objective has saturated and begun to overfit. We are explicit about limitations, and identify two directions kept as future work: extending RL to the discrete decisions of the vision stage (frame selection, landmark selection, decoding strategy), and a two-stage gloss branch.

Keywords: sign language translation, video-to-text translation, video representation learning, spatio-temporal modelling, pose estimation, reinforcement learning, SCST, exposure bias, PHOENIX-2014T, How2Sign

1 Introduction

Sign language is the primary means of communication for the Deaf community. Sign Language Translation (SLT) aims to map a sign-language video sequence $V = (v_1, \dots, v_T)$ to a text sentence $Y = (y_1, \dots, y_N)$. This is a cross-modal sequence-to-sequence task: the input is a continuous visual signal while the output is a discrete, grammatical token sequence in the target language [1].

SLT as a video-understanding problem.

It is tempting to read SLT as machine translation with an unusual input, but the difficulty is concentrated on the visual side. A PHOENIX-2014T sentence is a video of roughly 100 to 300 frames in which meaning is carried by the *joint* spatio-temporal configuration of two hands, the arms and the torso: the same handshape means different things depending on where it is placed and how it moves, so neither a single frame nor a bag of frames is sufficient. Before any language model can act, the pipeline must therefore solve a chain of classical image and video processing problems — detect and track the signer’s body and hands under motion blur and self-occlusion, recover a stable geometric representation per frame, handle frames where detection simply fails, normalise a variable-length stream to a fixed temporal budget, and finally encode a $[T, \text{joints}, \text{channels}]$ tensor into a memory that a decoder can attend over. Each of those steps is a modelling decision that changes the final BLEU, and we treat them as such. Our results bear this out: the single largest effect measured anywhere in this study is not the choice of RL algorithm but the choice of *visual representation*, where RGB-derived I3D video features outscore skeletal pose by more than a factor of two on How2Sign (Section 6).

SLT should be distinguished from two neighbouring tasks. Sign Language Recognition (SLR) maps video to glosses or labels, which is essentially classification and often uses CTC, while Sign Language Production (SLP) goes in the opposite direction, from text to motion. Of the three, SLT is where reinforcement learning is most natural, precisely because it is a grammatical sequence-generation problem, exactly where the two classic weaknesses of teacher-forcing training surface: exposure bias [2] and loss-metric mismatch.

This project serves two courses at once, Reinforcement Learning and Image and Video Processing, under a single guiding principle: do not separate the two directions. Every visual choice is paired with the question of what RL can exploit, and every RL design is rooted in a concrete vision pipeline. Concretely, the video stage supplies the state that the RL agent conditions on, so a weak visual representation caps the reward the agent can ever reach; conversely, the sequence-level reward is the only signal in the system that can eventually be pushed back into discrete visual decisions such as which frames to keep (Section 4.5).

The visual front-end and its cost.

For feasibility we use a pose-only 183-d-per-frame input extracted with MediaPipe Holistic [29] rather than raw RGB: it is computationally light, invariant to background and clothing, and feasible under a free GPU budget, which is a hard constraint of the project. Even so, it is a genuine video-processing stage in its own right, and the expensive one — landmark extraction over the full corpus costs 10 to 20 CPU-hours, an order of magnitude more wall-clock than any single training run, which is why it is decoupled and cached (Section 4). The cost of the choice is equally concrete: RGB-hungry video backbones such as I3D [28], SlowFast, TimeSformer and VideoMAE fall outside the trainable scope, so we quantify what that costs by running *precomputed* I3D features against pose features on How2Sign, turning the constraint into a measurement rather than an assumption.

This paper makes four contributions.

1. A controlled comparison of six interchangeable spatio-temporal video encoders behind a single interface (Experiment 4), spanning the three canonical inductive biases for skeletal video — global attention, convolutional temporal locality, and anatomical graph priors — reported jointly against parameter count and inference latency, and a modality comparison of RGB-derived I3D video features against skeletal pose (Experiment 13) that measures what the pose-only constraint costs.
2. A systematic study of eight RL decoder-fine-tuning algorithms and variants for SLT on small data, with a priority table grounded in a realistic compute budget, a pragmatic angle rarely discussed by state-of-the-art work.
3. A systematic reward ablation (Experiment 9), clarifying the trade-off between score and true quality such as repetition and sentence length.
4. A reproducible codebase on free hardware (Kaggle T4 x2), including the cached pose-extraction stage and an automatic comparison table with mandatory floor baselines, organised around a training-subset sweep (5, 10, 25%; full dev/test) that doubles as a data-size ablation, plus a How2Sign trial across two sign languages and two visual modalities.

We additionally sketch, as future work (Section 4.5), two directions built into the codebase design but not evaluated here: extending RL to the discrete decisions of the vision stage, and a two-stage gloss branch.

2 Related Work

2.1 Sign Language Translation

Camgoz et al. [1] defined SLT as distinct from SLR, proposed an attention-based sequence-to-sequence model, and introduced PHOENIX-2014T, which became the standard benchmark for continuous SLT. Their follow-up [3] combined a CTC objective on glosses with a Transformer decoder, raising BLEU-4 to about 21.8 and providing the closest reference to our pipeline. Recent work leverages large language models and vision-language pretraining: GFSLT-VLP [4] performs CLIP-style gloss-free pretraining, and Sign2GPT [5] uses a frozen large language model with adapters. SPOTER [6], a lightweight Transformer over normalised pose, is the closest reference to our compute regime.

2.2 Visual Representations for Sign Language Video

Because the input is video, the representation question precedes the translation question, and the field has answered it in three broadly different ways. The *appearance* route encodes raw RGB with a video backbone: I3D [28] inflates 2-D ImageNet filters into 3-D spatio-temporal kernels and remains the standard clip-level feature extractor for sign language, and it is the feature that How2Sign [30] ships precomputed, which is exactly why it is available to us as a comparison point despite our inability to train an RGB backbone. The *geometric* route discards pixels and keeps a skeleton: MediaPipe Holistic [29] runs cascaded per-frame detection and landmark regression for body, face and hands in real time on CPU, producing the fixed-dimensional joint stream we use; SPOTER [6] shows that a small Transformer over normalised landmarks is competitive at word level while being orders of magnitude cheaper than pixel models. Between them sits the *structural* route, which keeps the skeleton but models it as a graph rather than a flat vector: ST-GCN [22], introduced for skeleton-based action recognition, alternates graph convolution over anatomically connected joints with temporal convolution across frames, and is the direct ancestor of three of our six encoders.

The trade-off between these routes is the central video-processing question of this paper. Appearance features retain information that landmarks destroy — facial grammar, mouthing, hand-object and hand-body contact, and fine finger detail lost to occlusion or tracking failure — at a compute cost that a free-hardware budget cannot pay. Geometric features are compact, background invariant and cheap, but they are the output of an upstream detector that was never trained on signing, so every extraction error becomes silent, irrecoverable input noise. Recent gloss-free systems increasingly hedge by combining the two, for example RVLF [16], which fuses skeleton features with DINOv2 hand and face crops. We do not resolve the trade-off architecturally; instead we measure it directly (Experiment 13, Section 6), holding the decoder, reward and RL routine fixed while swapping only the visual features.

2.3 Reinforcement Learning for Sequence Generation

MIXER [2] first combined a cross-entropy warm start with REINFORCE for sequence-to-sequence models and identified exposure bias. SCST [7] uses the greedy decode itself as a baseline, removing the need for a separate critic, and is the backbone of our system. Other directions include MRT [8], which computes an expected risk over N candidates, RAML [9], which performs maximum likelihood on a noised target without rollout, and the modern policy-gradient family, GAE [10] and PPO [11]. DPO [12], which inherits the RLHF spirit of InstructGPT [13] but drops the PPO loop, recasts preference optimisation as classification over preference pairs. Crucially, Kiegeland and Kreutzer [14] argue that much of the apparent gain of RL for machine translation is illusory unless exploration and data scale are controlled, a caveat we respect in our ablation design (Section 5).

2.4 RL for SLT: Direct Precedents

Ordered by proximity to our setting, Panaro [15] applies SCST and PPO directly to RWTH-PHOENIX-Weather-2014T, making it the closest precedent by both dataset and algorithm, though it is a master’s thesis and not peer reviewed. Two recent preprints continue this line. RVLF [16] is, to our knowledge, the first work to apply GRPO to SLT: it combines skeleton features with DINOv2 hand and face features, performs instruction tuning of a large language model, and then optimises with GRPO under a reward that combines translation fidelity (BLEU) with sentence completeness (ROUGE) — a reward composition close in spirit to ours. It reports BLEU-4 gains of +5.1 on CSL-Daily, +1.11 on PHOENIX-2014T, +1.4 on How2Sign, and +1.61 on OpenASL. SignDPO [17] applies multi-level DPO to skeleton-based gloss-free SLT, generating non-preferred samples by hierarchical spatial-temporal perturbation and using decoder cross-attention to locate semantically salient skeletal regions; it is evaluated on CSL-Daily, How2Sign, and OpenASL, and notably *not* on PHOENIX-2014T, so its numbers are not directly comparable with ours.

Two distinctions position our work. First, both recent preprints operate at large-model scale — RVLF around a language model with DINOv2 features, SignDPO across three large benchmarks — whereas we deliberately target a small-compute, single-dataset regime on free hardware. Second, our positioning is systematic breadth (eight algorithms and variants, a reward ablation) under an explicit compute budget, rather than pushing a single state-of-the-art point; a further direction, unique among these precedents, is to extend RL to the discrete decisions of the vision stage, which we leave as future work (Section 4.5). Both 2025–2026 preprints should have their publication venue confirmed before formal citation.

3 Background and Problem Formulation

3.1 The SLT Problem

Given a pose-feature sequence $X = (x_1, \dots, x_T)$ with $x_t \in \mathbb{R}^{183}$, the goal is to learn the conditional distribution $p_\theta(Y | X)$ over the target sentence $Y = (y_1, \dots, y_N)$. The

encoder-decoder model factorises as

$$p_{\theta}(Y \mid X) = \prod_{t=1}^N p_{\theta}(y_t \mid y_{<t}, \text{Enc}(X)). \quad (1)$$

The final evaluation is sentence-level translation quality (BLEU-4 and its variants).

3.2 Why Cross-Entropy Is Not Enough

Teacher-forcing training minimises $\mathcal{L}_{\text{CE}}(\theta) = -\sum_t \log p_{\theta}(y_t^* \mid y_{<t}^*, X)$, where $y_{<t}^*$ is always the ground-truth token. This causes three problems.

- Exposure bias. During training the decoder always receives the correct token, but at inference it receives its own generated token, so errors accumulate along the sequence [2]. This is more severe in SLT because of the small dataset (about 7K sentences) and noisy visual input.
- Loss-metric mismatch. Training optimises token-level cross-entropy, but evaluation uses sentence-level BLEU, a non-differentiable function, so no gradient $\partial \text{BLEU} / \partial \theta$ exists.
- Degeneracy. The model favours common safe tokens and tends to repeat phrases [18], which is exactly why our reward has a repetition-penalty term rather than a superfluous option.

3.3 MDP Formulation

RL addresses all three by directly optimising the expected reward $J(\theta) = \mathbb{E}_{Y \sim p_{\theta}}[R(Y)]$, where R is a sentence-level metric. We cast decoding as a Markov Decision Process. The agent is the decoder π_{θ} , the sole decision maker; the encoder is not an agent and only produces observations. The environment is the encoder memory together with the reference sentence (fixed within an episode) and the reward scorer, with a deterministic transition $s_{t+1} = \text{concat}(s_t, a_t)$. The state is $s_t = (\text{Enc}(X), y_{<t})$, the action a_t is choosing a token, the reward is a sentence-level metric at the end of the episode (sparse), and an episode is one sentence from bos to eos. The policy gradient (REINFORCE [19]) with baseline b and advantage $A = R - b$ is

$$\nabla_{\theta} J(\theta) = \mathbb{E}[(R(Y) - b) \nabla_{\theta} \log \pi_{\theta}(Y)]. \quad (2)$$

Remark 1 Cross-entropy is a necessary condition, a warm start, because RL from scratch has too sparse a reward, but it is not sufficient. Hence a two-phase pipeline in which cross-entropy is followed by RL. RL creates no new linguistic information; it merely re-shapes the distribution learned from cross-entropy. Therefore every comparison must contrast cross-entropy only against cross-entropy plus RL from the same base checkpoint.

4 Methodology

Figure 1 summarises the full system. Data flows left to right through four shared stages (L1 to L4), which are warm-started with cross-entropy and then fine-tuned by the RL block (L5), converging at the evaluation stage (L8).

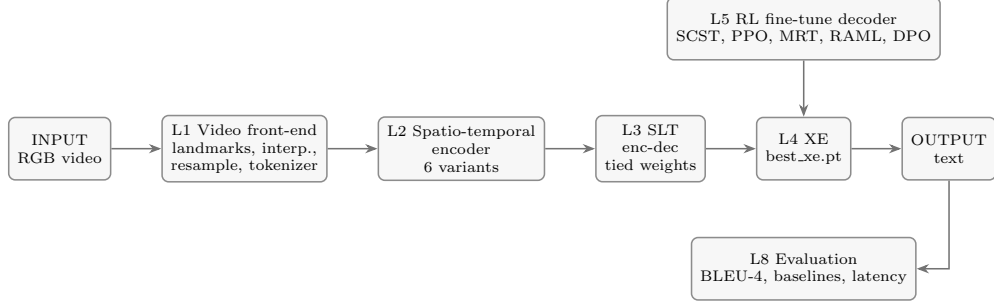


Fig. 1 The Sign-Language-REL architecture. The video front-end (L1) turns RGB frames into a $[T, 183]$ landmark tensor; L2 encodes it spatio-temporally (six interchangeable variants); L3 and L4 form the encoder-decoder warm-started with cross-entropy; the RL block (L5) then fine-tunes the decoder directly toward the metric. Evaluation is stage L8. Two further RL upgrade blocks (RL over the discrete vision decisions of L1, and a two-stage gloss branch) are kept as future work.

4.1 Video Front-End: Landmark Extraction and Temporal Preprocessing (L1)

L1 is the image and video processing stage of the system, and it converts a variable-length RGB video into the fixed-shape tensor $[T, 183]$ that everything downstream consumes. It has four distinct steps, each a modelling decision rather than a formality.

Per-frame landmark regression.

Every frame is passed through MediaPipe Holistic [29], a cascade that first localises the person, then crops and runs dedicated regressors for the body and for each hand. The result is projected to a 183-d frame vector: 33×3 body landmarks (x, y plus the visibility score) and 21×2 landmarks for each hand. Visibility is retained for the body, where it is informative about occlusion, and dropped for the hands, where the detector reports it unreliably; keeping an untrustworthy channel would only add noise across the 84 hand dimensions. This step reduces each frame from $\sim 10^5$ pixels to 183 numbers, which is the compression that makes the rest of the study affordable and, simultaneously, the point at which facial grammar and mouthing are irreversibly discarded (Section 2.2).

Handling detection failure.

Hand tracking fails routinely in signing video — fast motion blurs the hand, the hands cross or overlap the face, and one hand often leaves frame entirely — so a nontrivial fraction of frames come back with missing landmark groups. Zero-filling them would be

actively harmful, since a zero vector is a valid coordinate meaning “origin” and would inject a spurious jump toward the top-left corner of the image. We instead interpolate linearly *per segment* (body, left hand, right hand independently), so a dropped hand is bridged along its own trajectory while the successfully tracked groups in the same frame are left untouched.

Temporal normalisation.

Sequences are uniformly resampled or truncated to at most $T = 300$ frames. This bounds the $O(T^2)$ attention cost of the global encoders and equalises batch shapes, but it is deliberately the simplest possible policy: uniform sampling assumes every frame is equally informative, which is plainly false for signing video, where transitional movement between signs carries far less information than the sign nuclei. We keep it as a baseline precisely so that a learned frame-selection policy has an honest reference to beat (Section 4.5).

Text side.

Target text is tokenised with SentencePiece BPE, with a vocabulary size of 3000 and a maximum text length of 60.

Remark 2 Pose extraction is a CPU bottleneck of 10 to 20 hours, so it is fully decoupled from training: results are cached as npz files and uploaded as a Kaggle Dataset. Re-extracting on every training run would break the GPU budget.

4.2 Spatio-Temporal Video Encoders (L2)

L2 is where the video-representation question of Section 2.2 becomes an experiment. A pose sequence is a *graph in time*: within each frame the 75 tracked joints form a spatial skeleton, and across frames they trace trajectories. This is the same [time, space, channel] structure that a raw video tensor has, with the spatial axis reduced from a pixel grid to a set of tracked keypoints, so the design space is the familiar one of video representation learning — and the encoder must commit to an inductive bias about which of the two structures to exploit and how. We deliberately span the three canonical families of that design space rather than tune a single network, because the correct bias for skeletal SLT is an empirical question, not a settled one, and because a single interface makes it a controlled one. All six encoders implement the same signature $\text{Enc}: (\text{pose}, \text{mask}) \mapsto [B, T, d_{\text{model}}]$ behind one factory, so swapping is a single flag and the decoder, RL trainers and reward are byte-identical across them. This isolates the encoder as the only variable in Experiment 4.

The three inductive-bias families.

Table 1 lists the six encoders; they group as follows. *Global attention* (P1 Transformer, P4 Graph Transformer) imposes no locality: every position may attend to every other, letting the model discover long-range dependencies but paying an $O(T^2)$ cost per layer and needing more data to identify the useful connections. *Convolutional locality* (P5 TCN, and the temporal half of P3 ST-GCN) hard-codes the prior that nearby frames

matter most, buying linear cost and strong sample efficiency at the price of a bounded receptive field, widened here by exponential dilation. *Anatomical graph priors* (P2 GCN, P3 ST-GCN, and the learned-edge variant P4) inject the physical skeleton as a fixed or learned adjacency, so information flows along joints that are actually connected; this is the bias most specific to pose and the one we expect to help most when data is scarce. P6 Perceiver IO is orthogonal to all three: it is an *efficiency* strategy, a learned latent bottleneck of size $n_{\text{lat}} \ll T$ that reduces the dominant attention cost to $O(T n_{\text{lat}})$, linear in sequence length, which is the property that matters most for the 300-frame pose streams here.

Table 1 The six spatio-temporal encoder architectures. Parameter counts are measured on the project configuration; complexity is the dominant term in the sequence length T .

Code	Architecture	Params	Cost	Inductive bias / role
P1	Transformer	8.26M	$O(T^2)$	Global self-attention, the main baseline
P2	GCN	9.38M	$O(T)$	Fixed anatomical graph, spatial prior
P3	ST-GCN	6.33M	$O(T)$	Spatial graph + temporal Conv1D [22], lightest
P4	Graph Transformer	9.48M	$O(T^2)$	Learned joint edges, factored spatio-temporal attention
P5	TCN	7.47M	$O(T)$	Dilated temporal convolution, local, highly parallel
P6	Perceiver IO	9.59M	$O(T n_{\text{lat}})$	Latent-bottleneck cross-attention [23]

Concretely, P1 lifts each frame to d_{model} and applies a pre-norm Transformer over time. P2 replaces that projection with two graph-convolution layers over a hand-built joint adjacency before pooling into the temporal Transformer, so anatomy is fixed rather than learned. P3 interleaves spatial graph convolution with temporal 1-D convolution in the ST-GCN manner, the only design that keeps both structure and time in linear cost. P4 keeps the graph idea but *learns* the joint relations, running self-attention over the 75 joints within a frame (spatial) and then a Transformer over time (temporal), a factored spatio-temporal attention. P5 stacks dilated, non-causal 1-D convolutions to grow the temporal receptive field exponentially before a half-depth Transformer, on the local-then-global principle. P6 compresses the sequence into a small latent array, processes it with self-attention, and cross-attends back to the original T positions so the output shape is preserved.

How encoders are evaluated.

We report two axes rather than accuracy alone: end-task BLEU-4 (Experiment 4) and *accuracy per parameter*, since a lighter encoder that matches a heavier one is the more

useful result for deployment. Parameter count is the stable efficiency axis; wall-clock latency is reported (Experiment 15) but treated as secondary because it varies with the sequence-length distribution of each subset. Because the decoder, reward and RL routine are held fixed, any BLEU difference is attributable to the encoder bias alone.

Remark 3 Measured parameter counts show that GCN and Graph Transformer are *not* lighter than the Transformer at this configuration, contrary to the common assumption that graph models are cheap; only ST-GCN is genuinely lighter. ST-GCN drops the visibility channel, using only the x and y coordinates, so that all 75 joints share the same channel count, yielding an effective 150-d input rather than 183-d. This is a confound to note when interpreting Experiment 4, and it is why parameters, not an assumed complexity class, are used as the efficiency axis.

4.3 SLT Model and the Cross-Entropy Phase (L3 and L4)

The core model is a Transformer encoder-decoder [20] with a model dimension of 256, 4 heads, and 4 layers, deliberately small because the dataset is small and RL is very prone to overfitting. Output weights are tied to the token embedding [21] to reduce parameters. The model uses pre-layer-normalisation and sinusoidal, non-learned positional encodings, with label smoothing of 0.1 to retain entropy for exploration once RL begins. A value head, a critic $V(s)$, is attached for PPO and A2C. Phase one minimises teacher-forcing cross-entropy with mixed precision, a learning rate of $5\text{e-}4$ (2000-step warmup then cosine), weight decay of $1\text{e-}4$, gradient clipping of 1.0, up to 80 epochs with early-stop patience of 10. The result, the best checkpoint, is the warm start for every RL algorithm.

4.4 RL Decoder Fine-Tuning (L5) and Reward Design (L6)

Eight algorithms share a single reward formulation and are selected via one flag (Table 2). Every reward is used as an advantage against a baseline, the reward of the sample minus the reward of the greedy decode, following Equation 2, to avoid variance and difficulty bias. The general reward is

$$R = w_{\text{bleu}} \text{BLEU} - w_{\text{rep}} \text{rep} - w_{\text{len}} \text{len} (+ w_{\text{sem}} \text{sem}). \quad (3)$$

Setting a weight to zero disables that term, which is the infrastructure for the reward ablation (Experiment 9). Sentence-level BLEU uses effective order so that short sub-4-gram sentences, which are very common in PHOENIX, are not scored zero incorrectly; the repetition penalty is measured on tri-grams.

The self-critical policy gradient.

The estimator of Equation 2 is unbiased for any baseline b that does not depend on the sampled sequence, but its variance is dominated by the choice of b . We use the *self-critical* baseline of [7]: the model decodes twice from the same state, one sampled sequence $y^s \sim \pi_\theta$ and one greedy sequence $y^* = \arg \max \pi_\theta$, and the advantage is

their reward difference,

$$\nabla_{\theta} J(\theta) = - (R(y^s) - R(y^*)) \nabla_{\theta} \log \pi_{\theta}(y^s), \quad A = R(y^s) - R(y^*). \quad (4)$$

This has three consequences that motivate the whole L5 design. First, the baseline is the model’s own test-time decode, so it costs no extra parameters and no learned critic, and it automatically tracks the policy as it improves. Second, A is difficulty-invariant: an intrinsically hard sentence lowers both $R(y^s)$ and $R(y^*)$, so the gradient reflects only whether *this* sample beat the current greedy policy, not how hard the sentence was, which removes the per-sentence difficulty bias that makes absolute-reward REINFORCE unusable at this data scale. Third, because y^s is generated autoregressively exactly as at inference, the rollout distribution equals the test distribution, which is the direct structural fix for exposure bias. Figure 2 shows the resulting loop; the eight algorithms differ only in how the single advantage A is turned into a parameter update.

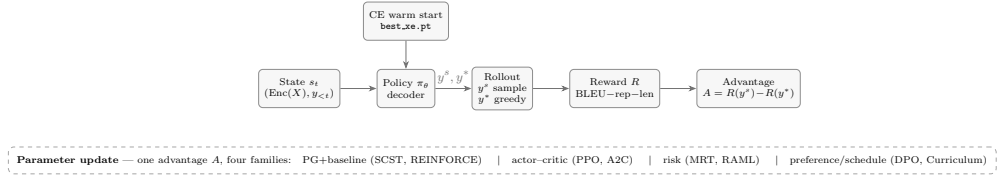


Fig. 2 The RL decoder fine-tuning loop (L5). The decoder is the policy; it produces both the action (y^s) and its own baseline (y^*). The reward (L6) is non-differentiable, so learning flows only through the advantage A . The eight algorithms are four families that reuse this same loop, differing only in the update rule applied to A .

Four algorithm families.

The eight methods (Table 2) are not eight unrelated tricks but four families spanning the standard axes of sequence-level RL, chosen so the comparison answers a specific question rather than simply enumerating baselines. *Policy gradient with a baseline* (SCST, REINFORCE) is the backbone: SCST is Equation 4, and REINFORCE is the same estimator with $b = 0$, included solely as the ablation that quantifies how much the self-critical baseline is worth. *Actor-critic with a trust region* (PPO, A2C) replaces the greedy baseline with a learned value head $V(s)$ and GAE advantages, and constrains the update with a clipped surrogate; A2C is the same without the clip, isolating the effect of the trust region. This family tests hypothesis H3, whether a heavier, more stable optimiser beats the cheap self-critical estimator; H3 is deliberately neutral, since either outcome is informative. *Expected-risk minimisation* (MRT, RAML) optimises over a distribution of candidates rather than one sample: MRT minimises risk over N re-normalised candidates, and RAML avoids rollout entirely by sampling reward-shaped targets, trading rollout cost for lower variance. *Preference and schedule methods* (DPO, Curriculum) sit outside the reward-gradient frame: DPO reduces alignment to logistic classification over self-generated (better, worse) pairs, and Curriculum shapes the reward and sentence-length ordering over epochs to soften the RL cold start.

Table 2 The eight RL algorithms grouped into four families, all sharing the reward of Equation 3 and the advantage of Equation 4.

Family	Algorithm	Mechanism	Source
PG + baseline	SCST	Baseline is the greedy decode, no critic	[7]
	REINFORCE	Baseline zero, high variance, ablation	[19]
Actor-critic	PPO	Clipped surrogate, GAE critic, multi-epoch	[11]
	A2C	Same without the clip, one epoch	[11]
Expected risk	MRT	Expected risk over N candidates	[8]
	RAML	ML on a reward-noised target, no rollout	[9]
Preference / schedule	DPO	Classification on self-generated pairs	[12]
	Curriculum	Reward ramp plus short-to-long ordering	

How algorithms are evaluated.

All eight share the identical warm start, encoder, reward and evaluation, so any BLEU difference is attributable to the update rule alone. The reference is not another RL method but the CE-only checkpoint (Section 6.6): an RL method is credited only when it beats CE by more than the observed run-to-run noise. Because RL variance is high and most runs use a single seed, we privilege trends that repeat across the four families and across the subset sweep over any single cell, and we watch three reward-hacking signals (BLEU up while ROUGE down, rising tri-gram repetition, advantage collapsing to zero) as falsification checks rather than assuming success.

Remark 4 The RL learning rate of $5e-6$, which is 100 times smaller than the cross-entropy rate, is the most sensitive hyperparameter of the system, standard in the SCST and RLHF literature. For PPO on sequence-to-sequence models, the encoder and hidden states must be recomputed with gradients at each PPO epoch; reusing memory from the no-gradient rollout would prevent the encoder from ever receiving gradients from the PPO loss, the most common implementation bug.

4.5 Future Extensions Built Into the Design (Not Evaluated Here)

The codebase is structured so that two further directions plug into the same backbone without altering the rest; both are left as future work and are *not* part of the reported study. First, extending RL to discrete decisions in the vision pipeline: a small policy could select frames (a Bernoulli keep/drop per frame, reward equal to BLEU from a frozen SLT model minus a kept-frame penalty), vary the temporal sampling rate with content, select landmark groups to handle occlusion, or choose a per-input decoding temperature. A key caveat motivates deferring this: a soft mask that merely zeroes dropped frames leaves the sequence length unchanged, so it would measure which

frames matter but would not prove reduced compute; genuine savings require re-indexing the sequence before the encoder. Second, a two-stage gloss branch (pose-to-gloss via CTC, then gloss-to-text via a small NMT model) reusing the L2 encoder, which would need the gloss annotation axis and doubles training. Both directions are documented in the repository.

5 Experimental Setup

5.1 Dataset

The primary dataset is PHOENIX-2014T [1], continuous SLT for German Sign Language, with 8257 sentences split as 7096/519/642 for train, dev, and test. It is the only dataset that simultaneously has reference sentences for reward, is small enough for many RL runs within budget, and has no access barrier. The 183-d poses are self-extracted and cached. To control compute, the training split is subsampled with a fixed seed of 42, while the development and test sets are always evaluated in full. The study is organised around three nested training-subset sizes (5, 10, and 25%; the nesting under a fixed seed makes cross-subset comparison controlled) that together constitute the data-size ablation (Experiment 11); still larger subsets (50%, 100%) are supported by the same code when quota allows. Because the number of epochs is fixed across subsets, training time scales roughly with the amount of training data, but the full-size dev/test evaluation does not shrink, so smaller subsets are not proportionally faster. The second dataset (Experiment 13) is How2Sign [30], American Sign Language, run with the same encoder-decoder plus CE-then-SCST pipeline at the 10, 25, and 50% subsets. It plays two roles. As a *modality* experiment it is the only place in the study where we can compare visual representations at the top of the pipeline, because How2Sign ships precomputed I3D video features (1024-d, RGB-derived, from an inflated 3-D CNN [28]) alongside our own MediaPipe body-pose features (99-d); swapping the feature file while holding the decoder, reward and RL routine byte-identical isolates the visual representation as the only variable, and is our measurement of what the pose-only constraint costs. As a *generalisation* experiment it changes the sign language and, importantly for the RL question, the target-length regime: How2Sign sentences average 17.4 words against 13.96 for PHOENIX, with a far heavier tail (90th percentile 33 against 22 words, maximum 195 against 52), so a decoded episode is substantially longer and exposure bias has more steps over which to compound. How2Sign is trained and reported separately from PHOENIX rather than merged into the same ranking, since its far lower BLEU baseline makes the reward sparse and a null result still informative.

5.2 Evaluation Metrics

The translation-quality group is BLEU-1 to BLEU-4 [24], where BLEU-4 is the headline but BLEU-1 to BLEU-3 are always reported because PHOENIX sentences are short, together with ROUGE-L, METEOR, and, at the final stage, BERTScore [25] and COMET [26]. BLEU is computed with sacreBLEU [27]. The system-performance group is latency at batch sizes 1 and 16 with warm-up, throughput, peak memory,

and parameter count. BERTScore and COMET are used only at final evaluation, not inside the per-step reward loop, because of their GPU cost.

Remark 5 BLEU-4 should not be the sole early-stopping criterion in advanced RL experiments; the n-gram repetition rate and mean length should also be monitored to detect reward hacking that BLEU alone can be fooled by. When comparing two close configurations such as PPO against SCST, run at least two seeds, since RL has high variance.

5.3 Mandatory Floor Baselines

Every number must be read relative to model-free floors (Table 3), generated by a baseline script and automatically merged into the comparison table. In addition, the cross-entropy-only checkpoint (xe) is the reference against which every RL algorithm is measured, since RL creates no new linguistic information.

Table 3 Model-free floor baselines. Beating the floor is necessary for any number to be meaningful.

Baseline	Controls for and rationale
base_empty	The absolute floor of BLEU: an empty hypothesis for every sentence.
base_most_frequent	The single most frequent training sentence, emitted for every test input. PHOENIX repeats a lot, so this already scores non-trivial BLEU with no model, and a model that does not clear it has a meaningless score.

5.4 Implementation Details

Table 4 lists the main hyperparameters. Hardware is Kaggle T4 x2 with a budget of about 30 GPU-hours per week. The full experiment matrix (six encoders, eight RL methods, reward ablation, latency) costs roughly 6 to 9 hours at the 5% subset, which fits comfortably within one session; the 25% matrix is about 20 to 25 hours, and larger subsets are spread across multiple sessions. The orchestrator is resumable, so an interrupted session resumes from completed steps.

5.5 Experiment Suite

Table 5 summarises the feasible experiments retained in this study. Experiments needing modalities or branches outside the current scope (RGB input; the RL-beyond-decoder frame/landmark/decode policies; the gloss two-stage branch; cross-dataset transfer) are deferred to future work; the original experiment numbering is kept so that cross-references do not shift. Each experiment maps directly to a command-line flag.

Five research questions guide the design, and the first two are vision questions. RQ0 asks how much the visual representation itself is worth: what does the pose-only

Table 4 Main hyperparameters from the project configuration.

Parameter	Value	Parameter	Value
pose dimension	183	model dimension	256
max frames	300	heads and layers	4 and 4
vocabulary size	3000	dropout	0.3
max text length	60	label smoothing	0.1
xe learning rate	5e-4	xe epochs	80, patience 10
rl learning rate	5e-6	rl epochs	20
rl batch size	8	gradient clip	1.0
GAE lambda	0.95	reward weights	1, 0.5, 0

Table 5 The retained experiments and their priority.

No.	Content	Priority
1	Cross-entropy against SCST	Mandatory
9	Reward ablation, weight combinations	Mandatory
3	Single against multi-sample SCST, N in 1, 4, 8	High
7	PPO against SCST, plus MRT, RAML, DPO, A2C	High
4	Six spatio-temporal encoders, isolated variable	High
11	Data-size ablation of BLEU gain against subset	High
2	BLEU against BLEU plus penalty	Medium
10	Decoder ablation of layers and model dimension	Medium
15	Inference latency across six encoders, xe against rl	Medium
12	Generalisation gap, train minus test BLEU	Medium
13	How2Sign: visual modality (I3D video vs. pose) and cross-language, 10, 25, 50%	Medium
14	Few-shot, subset 0.01 and 0.02	Low

constraint cost relative to RGB-derived video features, with everything downstream held fixed? RQ4 asks which spatio-temporal inductive bias — global attention, convolutional temporal locality, or an anatomical graph prior — best encodes a landmark video, and how that interacts with the ability to benefit from RL. The remaining three are RL questions. RQ1 asks whether SCST improves over cross-entropy on small data, consistently across subsets. RQ2 asks how reward design affects true quality such as repetition and length. RQ3 asks whether PPO is worth the added complexity under a sparse end-of-sentence reward, in the light of [14].

Rather than the full 6×8 cross-product of encoders and RL methods (which would be forty-eight training runs), we vary one axis at a time. On the *encoder axis* (Experiment 4) we fix the algorithm to SCST and sweep all six encoders, so every encoder receives RL decoder fine-tuning. On the *algorithm and reward axis* (Experiments 7 and 9) we fix the encoder to the Transformer and sweep the remaining RL methods and reward variants. Table 6 summarises this coverage; the Transformer is the only encoder that participates in both axes.

Table 6 Coverage of the two experimental axes. Every encoder is fine-tuned with SCST (encoder axis); the additional RL methods and reward ablation are run on the Transformer only (algorithm axis).

Encoder	XE	SCST	PPO/MRT/RAML/DPO	REINFORCE/A2C/Curr.	Reward abl.
P1 Transformer	✓	✓	✓	✓	✓
P2 GCN	✓	✓			
P3 ST-GCN	✓	✓			
P4 Graph Transformer	✓	✓			
P5 TCN	✓	✓			
P6 Perceiver IO	✓	✓			

6 Results and Discussion

All numbers below are test-set BLEU-4 computed with sacreBLEU on the full 642-sentence PHOENIX test split, regardless of the training-subset size, and are aggregated automatically by `make_overview.py` from the per-run `test_results.json` files. Every number is read relative to the model-free floor (Table 3: empty 0.00, most-frequent 0.19) and, for the RL algorithms, relative to the cross-entropy checkpoint from the *same* base, since RL creates no new linguistic information. Because the GPU budget forces mostly single-seed runs, we lead with *trends that repeat across three subsets and many methods* rather than with any single cell, and we quantify the noise floor explicitly.

6.1 Cross-Entropy against RL, and the Data-Size Ablation

Table 7 is the headline result: the eight decoder fine-tuning algorithms against the cross-entropy baseline, on the Transformer core, across the three training subsets. This single table answers RQ1 (does RL beat cross-entropy?) and, read left to right, Experiment 11 (how the gain scales with data).

At the 5% subset the picture is the flat, honest null we reported previously: SCST and A2C tie for best at 4.31 against the cross-entropy 4.16, a +0.15 gap that is *smaller than run-to-run variance*. An independent re-run of the same SCST configuration inside the reward ablation reached 3.75 (Table 8, first-vs-default rows), a 0.56 swing that exceeds every algorithmic difference at 5%. So at the smallest subset we claim no reliable RL gain, consistent with the deliberately neutral H3 and with [14].

Table 7 Main results on PHOENIX-2014T: test BLEU-4 of cross-entropy against eight RL decoder fine-tuning algorithms (Transformer core), across the 5/10/25% training subsets (full dev/test). Δ is the gain over cross-entropy at the same subset. Best RL per subset in bold.

Configuration	5%	10%	25%
Floor, empty / most-frequent	0.00 / 0.19	0.00 / 0.19	0.00 / 0.19
Cross-entropy only (XE)	4.16	5.04	5.86
CE + SCST	4.31	5.23	5.94
CE + PPO	4.19	5.28	6.31
CE + A2C	4.31	5.14	6.03
CE + MRT	4.08	5.80	6.49
CE + RAML	3.99	5.37	6.30
CE + DPO	4.16	5.23	6.03
CE + REINFORCE (no baseline)	3.91	5.68	6.05
CE + Curriculum RL	3.77	5.16	5.76
Best Δ (best RL – XE)	+0.15	+0.76	+0.63

The data-size ablation is where the story changes, and it changes against our own prior. Moving to 10% and 25%, most RL methods now clear the cross-entropy baseline, and the *best* gain grows from +0.15 at 5% to +0.76 at 10% and +0.63 at 25% (Figure 3) — well outside the ≈ 0.5 noise band that swamped the 5% differences. This inverts hypothesis H1, which predicted a *larger* RL gain at smaller subsets on the grounds that exposure bias is worse there. Empirically the opposite holds: RL needs enough data to estimate a stable advantage before it can help, exactly the scale-dependence that [14] warn about. Two further details sharpen the reading. First, plain SCST is *not* what carries the gain; its own Δ stays modest (+0.15/ +0.19/ +0.08), and the leaders at 10 and 25% are the multi-sample estimator MRT (5.80, 6.49; Δ up to +0.76) and PPO (5.28, 6.31; Δ rising +0.03 $\rightarrow$ +0.24 $\rightarrow$ +0.45), whose advantage estimates are richer than SCST’s single greedy baseline and evidently benefit more from data. Second, PPO’s monotonically rising Δ is the one result that pushes gently against H3: with more data its learned critic becomes useful, though even at 25% it does not overtake MRT, so we do not upgrade H3 to a positive claim on a single seed.

6.2 Reward Ablation

Table 8 is the central table of the RL chapter, the reward ablation of Experiment 9, now run at all three subsets. Each row is a reward-weight combination on SCST; the question is whether the repetition and length penalties buy quality or merely cost BLEU.

The verdict is consistent across all three subsets and refutes H2 at this scale: the plain BLEU-only reward is the strongest everywhere (3.98/5.19/5.96), and every added penalty *lowers* BLEU, with the combined rep+len reward collapsing badly at the larger subsets (3.85 at 10%, 4.44 at 25%). The mechanism is visible in the 5% diagnostics: measured tri-gram repetition is already ≈ 0 under a pure-BLEU reward, so there is simply no reward-hacking for the repetition penalty to prevent, while the length

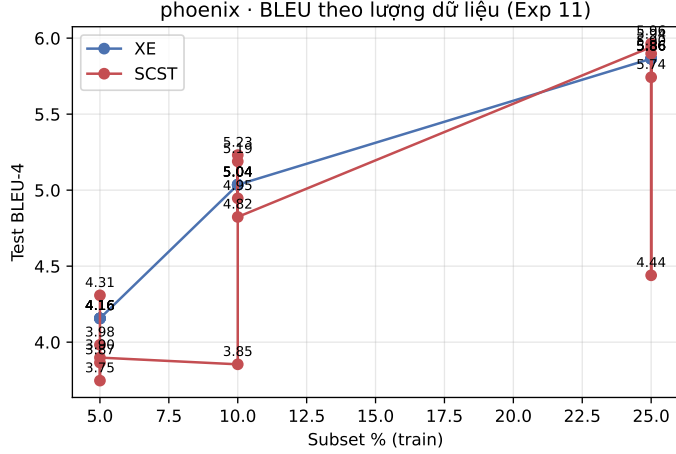


Fig. 3 Data-size ablation (Experiment 11). Best-RL and cross-entropy test BLEU-4 against training-subset size; the RL–CE gap is negligible at 5% and opens up at 10 and 25%, the opposite of the exposure-bias intuition in H1.

Table 8 Reward ablation of Experiment 9: SCST test BLEU-4 for four reward-weight combinations, across subsets. The 5% column also lists the measured tri-gram repetition rate and length ratio.

$w_{\text{bleu}}, w_{\text{rep}}, w_{\text{len}}$	Variant	5%	10%	25%	Rep.% (5%)	Len (5%)
1, 0, 0	BLEU only	3.98	5.19	5.96	0.0	0.113
1, 0.5, 0	BLEU – rep	3.75	4.82	5.90	0.0	0.077
1, 0, 0.5	BLEU – len	3.90	4.95	5.74	0.1	0.029
1, 0.5, 0.5	BLEU – rep – len	3.87	3.85	4.44	0.0	0.043

penalty shortens an output that is *already* far too short (length ratio $0.11 \rightarrow 0.03$) and thereby destroys n-gram matches. H2 assumed a degeneracy regime (repetition to be suppressed) that this small-data, brevity-limited system never enters; the penalties are pure cost here. This does not retire the penalty terms in general — they are the correct instrument once a model over-generates or repeats — but it says clearly that they should be *off* in this regime, and it is exactly the kind of finding the ablation was built to surface.

6.3 Spatio-Temporal Encoder Architecture and Efficiency

Table 9 serves Experiments 4 and 15 and is the video-representation result on the PHOENIX side: the six spatio-temporal encoders, decoder fixed, fine-tuned with SCST, across subsets, with the stable efficiency axis (parameter count) and a representative latency/throughput point. Since the decoder, reward and RL routine are byte-identical across rows, every difference here is attributable to how the encoder

models the [time, joints, channels] tensor. Recall the ST-GCN confound of an effective 150-d input.

Table 9 Six-encoder comparison (Experiments 4 and 15): SCST test BLEU-4 across subsets, with measured parameters and batch-1 latency / throughput at the 25% subset. Best BLEU-4 per subset in bold.

Encoder	Params	5%	10%	25%	Lat. / thr. (25%)
P1 Transformer	8.26M	4.31	5.23	5.94	115.9ms / 8.6
P2 GCN	9.38M	4.10	5.55	7.27	61.3ms / 16.3
P3 ST-GCN	6.33M	3.62	4.99	6.74	27.6ms / 36.3
P4 Graph Transformer	9.48M	2.66	5.53	7.64	115.6ms / 8.7
P5 TCN	7.47M	5.40	6.00	7.47	123.9ms / 8.1
P6 Perceiver IO	9.59M	4.08	4.77	6.54	104.5ms / 9.6

The most striking result of the study is that the encoder ranking *inverts* with data (Figure 4). At 5% the TCN is best (5.40) and the graph encoders are worst — Graph Transformer dead last at 2.66 — so the 5%-only reading refuted H4. But the graph encoders scale far better: at 25% the Graph Transformer (7.64) and GCN (7.27) are the two best, while the plain Transformer (5.94) is now the *worst* of the six. ST-GCN, the lightest at 6.33M, climbs 3.62 → 4.99 → 6.74 and overtakes the Transformer at 25% (6.74 against 5.94) with 23% fewer parameters — so H4, false at 5%, is actually *confirmed at scale*: a graph inductive bias does buy Transformer-level quality more cheaply, once there is enough data to train it. The TCN remains the best all-rounder, top or near-top at every subset (5.40/6.00/7.47). The methodological lesson is that architectural conclusions drawn at a single small subset can reverse entirely, and single-seed variance compounds this, so the ranking should be read as a data-dependent trend, not a fixed order.

On efficiency, parameter count is the stable axis and confirms the Section 4 remark: GCN and Graph Transformer are *not* lighter than the Transformer; only ST-GCN genuinely is. The latency/throughput numbers vary substantially across subsets because they depend on the sampled sequence-length mix rather than on architecture alone, so we report a single representative point and caution against over-reading it; at 25% ST-GCN is both the lightest and the fastest (27.6ms, 36.3 sent/s), which is architecturally consistent, making it the best quality-per-parameter and quality-per-millisecond choice at scale.

6.4 How2Sign: Visual Modality and Longer Targets

Experiment 13 does two things at once: it compares the visual representation at the top of the pipeline (RGB-derived I3D video features against skeletal MediaPipe pose features), and it asks whether the RL picture survives a different sign language and a longer target-length regime. It is run and reported separately from PHOENIX, never merged into the same ranking. Both features drive the identical CE-then-SCST recipe at the 10, 25, and 50% subsets (Table 10).

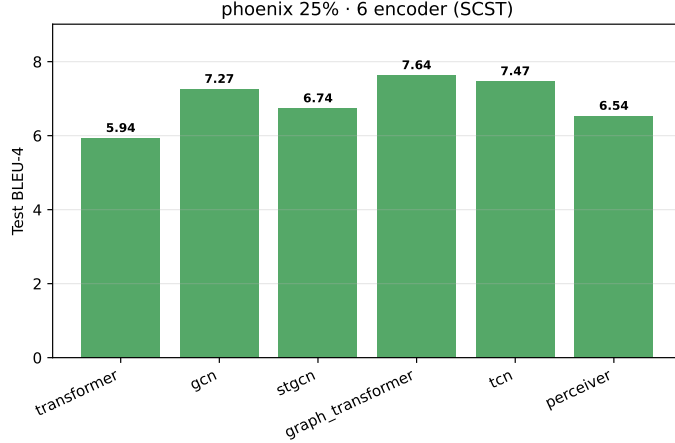


Fig. 4 Encoder comparison at the 25% subset (Experiment 4). Graph-based encoders (Graph Transformer, GCN, ST-GCN) overtake the plain Transformer, reversing the 5% ranking.

Table 10 How2Sign modality and generalisation trial (Experiment 13): test BLEU-4, cross-entropy against SCST, for RGB-derived I3D video features against skeletal MediaPipe pose features, across subsets. Decoder, reward and RL routine are identical across rows, so the I3D–MediaPipe gap is attributable to the visual representation alone. Reported separately from PHOENIX.

Feature	Metric	10%	25%	50%
I3D (1024-d)	XE	2.11	3.28	4.25
	SCST	2.22	3.52	4.35
MediaPipe (99-d)	XE	1.67	1.76	1.62
	SCST	1.59	1.93	1.25

The visual representation is the dominant factor.

The largest single effect measured anywhere in this study is the modality gap, and it is a video-processing result rather than an RL one. RGB-derived I3D features beat skeletal body pose at every subset (2.11 vs 1.67, 3.28 vs 1.76, 4.25 vs 1.62), and — more damaging for the pose route — only I3D *scales* with data (2.11 $\rightarrow$ 3.28 $\rightarrow$ 4.25) while pose stays flat around 1.6–1.9 and is, if anything, non-monotonic. Pose does not merely start lower; it stops responding to additional training data, which is the signature of an information ceiling in the representation rather than of an under-trained model. The interpretation follows directly from Section 4.1: the 3-D CNN retains appearance cues that landmark regression deletes by construction — facial grammar and mouthing, hand–body contact, fine finger configuration — and, unlike the landmark cascade, it degrades gracefully under the motion blur and self-occlusion that make hand tracking fail. This makes the pose-only constraint a measured ceiling

rather than a free design choice, and it is the strongest justification in the paper for the RGB and hybrid-feature directions listed as future work. It also reframes the RL contribution honestly: on this data, changing what the decoder *sees* is worth roughly 2.6 BLEU-4, whereas changing how the decoder is *optimised* is worth a few tenths.

Longer targets: where the sequence-level objective earns its place.

How2Sign is also the longer-sentence regime (17.4 words on average against PHOENIX’s 13.96, 90th percentile 33 against 22), and this is where the difference between a token-level and a sequence-level objective becomes visible in the training dynamics rather than only in a final number. Figure 5 plots dev BLEU-4 per epoch at the 10% subset for both modalities, with the SCST branch forking off the cross-entropy curve at its best checkpoint. The pattern is the same in both panels and is the point of the figure: *continued* cross-entropy training past the warm start does not merely stop helping, it actively loses ground, while the RL branch from the same checkpoint holds a flat, higher plateau. For I3D, cross-entropy peaks at 2.23 (epoch 18) and then oscillates down to a 2.08 mean over the remaining epochs ($0.93\times$ the peak), whereas the SCST branch averages 2.24 and reaches 2.29. For pose the contrast is starker still: cross-entropy peaks early at 1.70 (epoch 4) and collapses to a 1.13 mean ($0.66\times$ the peak, with individual epochs down to 0.65), while SCST holds 1.66 on average and peaks at 1.79. The same measurement on PHOENIX gives a milder version of the same effect (post-peak cross-entropy retention $0.79\text{--}0.89\times$ against $0.94\text{--}1.06\times$ for the RL branch), which is consistent with the exposure-bias account: the longer the target, the more decoding steps over which teacher-forced training and free-running inference can diverge, and the more a sentence-level reward computed on the model’s own roll-out has to correct. Under a fixed epoch budget, this is the practical value of the RL phase on long-target data — not a large jump in the best achievable score, but continued improvement in a regime where the token-level objective has already turned into overfitting.

Two caveats on that reading.

First, the dev-curve advantage does not fully convert into test BLEU-4. It does for I3D, where SCST beats cross-entropy at every subset ($+0.11/ +0.24/ +0.10$), but for pose the test deltas are inconsistent ($-0.08/ +0.17/ -0.37$) even though the dev curve favours SCST at 10 and 25%: a reward built on near-floor BLEU is too sparse to yield a reliably transferable gain, and the cross-entropy checkpoint being selected at its dev peak is a strong reference to beat. So Figure 5 should be read as evidence about *training dynamics* — what the two objectives do after the warm start — not as a claim of a guaranteed test-set win. Second, the How2Sign trial varies sign language, visual modality and target length simultaneously, so we present sentence length as the motivating observation and the divergence in the curves as the measurement, without claiming length is the isolated cause; separating them would need a length-stratified evaluation on a single corpus, which we did not run.

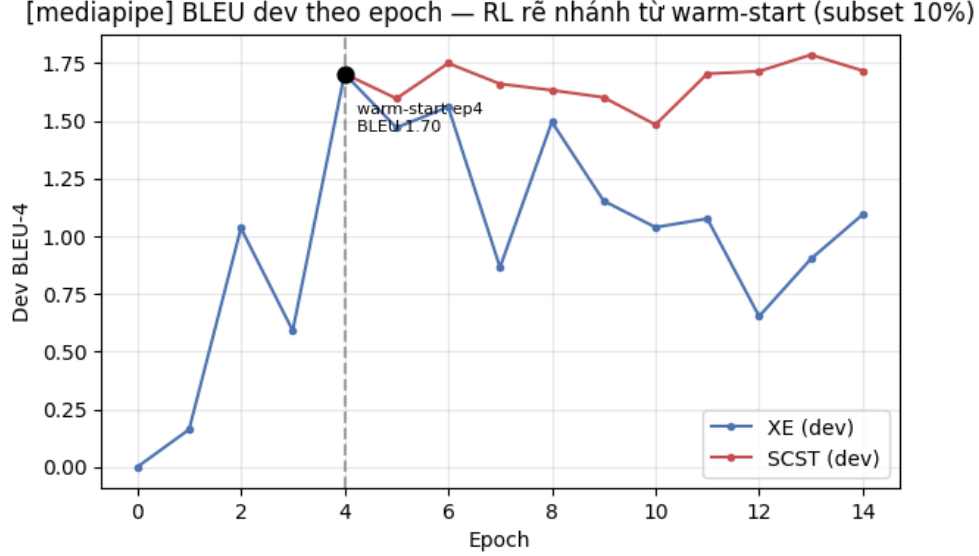


Fig. 5 How2Sign at the 10% subset: dev BLEU-4 per epoch, with the SCST branch (red) forking from the cross-entropy curve (blue) at its best checkpoint. Left, RGB-derived I3D video features; right, MediaPipe pose features. In both panels continued cross-entropy training degrades after the warm start while the RL branch holds a higher, flatter plateau — the effect is sharpest for the weaker pose signal, which overfits within four epochs. Note the vertical scales: the modality gap between the two panels is larger than any RL effect within either.

What reproduces.

Setting aside the caveats, the qualitative PHOENIX conclusion survives the change of language and modality: RL decoder fine-tuning yields a marginal, data-dependent improvement on top of a cross-entropy warm start, and it is most useful where cross-entropy saturates earliest. That this holds across two sign languages and two visual representations is stronger evidence than any single PHOENIX cell.

6.5 Hypotheses Revisited

We stated four hypotheses in advance (Section 5) and report each verdict honestly, including where the data contradicted us. **H1** (RL beats CE, more so at smaller subsets): *half right, half inverted*. RL does beat CE, but the gain *grows* with data ($+0.15 \rightarrow +0.76 \rightarrow +0.63$) rather than shrinking; exposure bias being worse at small data did not translate into a larger RL gain, because the advantage signal is too noisy there. **H2** (pure-BLEU reward hacks via repetition): *not supported at this scale*; measured repetition is ≈ 0 , the system under-generates, and every penalty only costs BLEU. **H3** (PPO need not beat SCST, kept neutral): *upheld*; PPO improves with data and its Δ rises to $+0.45$ at 25%, but it never overtakes MRT, so on a single seed we keep the neutral verdict. **H4** (ST-GCN matches the Transformer with fewer parameters): *false at 5%, confirmed at 25%*; ST-GCN overtakes the Transformer at

the largest subset (6.74 vs 5.94) with 23% fewer parameters, so the graph-inductive-bias argument holds once there is enough data. The recurring theme is scale: three of the four hypotheses only resolve correctly when read across the subset sweep, which is the strongest justification for having built the study around it.

6.6 Limitations

We are explicit about limitations so that results are not overstated. The subsets reported are small (5–25% of a corpus that is itself only $\sim 7\text{K}$ sentences), so absolute BLEU stays low (single digits, against ~ 9.5 for full-data pose-only and ~ 21.8 for gloss-based full-data [3]) and the data-size trends, while consistent, are read as directional rather than definitive. Most runs use a single seed because of the GPU budget, and RL variance is high (≈ 0.5 BLEU-4), so cross-method differences within a subset are interpreted cautiously and we lean on trends that repeat across subsets and methods. There is no human evaluation, so all conclusions rest on automatic metrics. The visual front-end carries the heaviest limitations. The input is landmark-only on PHOENIX, dropping the face entirely and therefore the mouthing and facial grammar that carry syntactic and adverbial content in sign languages; training an RGB backbone is out of scope on this budget, so the modality comparison relies on *pre-computed* I3D features that we could not fine-tune, which if anything understates the appearance route. Landmark quality is itself an unmeasured variable: MediaPipe was not trained on signing, we do not report detection-failure rates per corpus, and our per-segment interpolation silently smooths over whatever it missed. The temporal policy is uniform resampling to 300 frames, so long sequences lose frames non-selectively and the transitional-versus-nucleus distinction is ignored; a learned selection policy is exactly the deferred direction of Section 4.5. There is no true RLHF, since all RL uses automatic metrics as reward. Finally, the gloss-free setting is primary, so comparison with gloss-based methods in the literature is not fully fair; a two-stage gloss branch is left as future work (Section 4.5).

7 Conclusion

We presented a gloss-free sign language video-to-text pipeline, treating SLT as a video-understanding problem end to end: a landmark-based visual front-end (per-frame regression, per-segment interpolation of failed detections, uniform temporal resampling), a comparison of six spatio-temporal encoders behind one interface, and a decoder that is warm-started with cross-entropy and then fine-tuned by RL directly toward the metric, addressing the exposure bias and loss-metric mismatch of teacher-forcing training. On top of that sit a systematic comparison of eight decoder fine-tuning algorithms and variants under a shared multi-objective reward and a reward ablation separating BLEU gains from genuine quality, organised around a 5/10/25% training-subset sweep within a free-hardware budget, with a How2Sign trial across two sign languages and two visual modalities.

The clearest empirical result is a vision result, not an RL one: on How2Sign, RGB-derived I3D video features beat skeletal pose at every subset (4.25 against 1.62 BLEU-4 at 50%) and only I3D improves with more data, so the pose-only constraint is

a measured information ceiling rather than a free design choice — changing what the decoder sees is worth an order of magnitude more than changing how it is optimised. Within the pose representation, the encoder ranking inverts with scale: graph encoders and the lighter ST-GCN overtake the plain Transformer at 25%, confirming at scale the graph-inductive-bias hypothesis that failed at 5%, and ST-GCN is simultaneously the lightest (6.33M), the fastest (27.6ms per sentence) and better than the Transformer at that subset.

The second headline is a scale effect on the RL side. At the smallest 5% subset RL gives no reliable gain over cross-entropy — the SCST advantage is inside the noise — but a real, growing gain emerges at 10 and 25% (best Δ up to +0.76 BLEU-4), carried by the multi-sample MRT and by PPO rather than by plain SCST, which inverts the intuition that RL should help most when data is scarcest. The reward ablation shows that, in this brevity-limited regime, added repetition and length penalties only cost BLEU, since there is no degeneracy to suppress. On How2Sign, whose targets are markedly longer, the dev curves make the role of the sequence-level objective visible: continued cross-entropy training degrades after the warm start (retaining $0.66\text{--}0.93\times$ its peak) while the RL branch from the same checkpoint holds a flat, higher plateau — though for the weak pose signal that dynamic advantage does not fully convert into test BLEU, which we report as such. Methodologically we stress honest evaluation: every number is read relative to mandatory floor baselines, hypothesis H3 is deliberately neutral, three of four hypotheses are reported as partly or wholly contradicted by the data, and limitations are stated rather than hidden. Everything is designed to be reproducible on free hardware.

Several directions follow directly. First, and most distinctive, extending RL to the discrete decisions of the vision pipeline (frame, landmark, and decoding-strategy policies), with true sequence re-indexing so that frame selection measures real compute savings rather than only which frames matter. Second, a two-stage gloss branch (pose-to-gloss-to-text). Third, true RLHF with small-scale human preferences. Fourth, DPO with pseudo-preferences from multi-sample ranking, extending the perturbation idea of SignDPO [17]. Fifth, a video encoder to a query former to a small language model bridge. Sixth, enabling face landmarks. Seventh, cross-dataset transfer to a typologically different sign language.

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Declarations

- Funding: no funding was received for this work.
- Conflict of interest: the authors declare no competing interests.
- Data availability: PHOENIX-2014T is publicly available under its licence.
- Code availability: the codebase is designed to be reproducible on Kaggle T4 x2.

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